

EmoEcho: Designing Emotion Mimicry Mechanics for Enhancing Social Engagement in Digital Games

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ABSTRACT

Social interaction plays a critical role in enhancing player experiences in bodily play. Although previous research has extensively explored physical mimicry in movement-based games for social engagement, the incorporation of emotional expressions, a significant aspect of bodily interaction, remains underexplored. This study aims to broaden the understanding of bodily play by integrating emotion mimicry, examining its impact on enhancing social connections. We present “EmoEcho”, a two-player 2D side-scrolling game that embeds emotion mimicry into its core gameplay. EmoEcho allows players to dynamically alter the game environment using facial expressions, fostering a unique shared interactive experience. A pilot study compares emotion-based and keyboard inputs, revealing that emotion input significantly enhances the social dynamics of gameplay. This paper contributes to digital play by highlighting the potential of emotion mimicry in enhancing social connections, paving the way for more emotionally engaging and socially resonant gaming experiences.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design.**

KEYWORDS

Emotion Input, Games, Social Games, Facial Expression

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1 INTRODUCTION

In bodily play, the social dimension is crucial for enriching player experiences and fostering connections [6, 36]. Prior research in movement-based games has proposed numerous design strategies aimed at enhancing social engagement [25, 26]. One such strategy is body mimicry, where the game system encourages players to mirror each other’s physical interactions, thereby fostering a socially rich play experience [27]. While considerable work has been done on movement mimicry, it is important to note that bodily expression is not limited to mere movement: it also includes other aspects such as emotion [7]. This work aims to broaden the scope by delving into emotion mimicry, investigating its role in augmenting the social aspects of bodily play.

We believe that using emotion mimicry as a design resource in bodily play can lead to engaging experiences and foster deeper social connections. It is not just about copying movements but also about sharing and responding to emotional states, allowing players to communicate and empathize with each other in a way that aligns closely with human social interaction [13]. Psychological research underscores the role of emotion mimicry in building rapport and facilitating emotional synchrony among individuals, highlighting its societal significance [29]. Therefore, we suggest that emotion mimicry has the potential to create a shared emotional space that enhances the sense of togetherness in digital play. Our research delves into these possibilities, exploring the impact of emotion mimicry on social digital play.

To illustrate this concept, we introduce “EmoEcho”, a two-player 2D side-scrolling game that embeds emotion mimicry into the gameplay. This game allows players to modify the gaming environment in real-time using their facial expressions, creating an engaging and socially interactive space. In EmoEcho, Player A’s emotional expressions are translated into various in-game elements like obstacles and power-ups. Player B is then tasked with mimicking these expressions to interact effectively with these elements. A pilot study involving two pairs of participants was conducted. We set up a comparative study with two conditions: using emotion as input and using a keyboard as input, while keeping the gameplay identical in both scenarios. Participants experienced each condition and then completed the Game Experience Questionnaire (GEQ) core module and the social presence module [17]. The results indicated that the emotion input condition generally offered a better play experience,

with a notably higher score in social aspects. Semi-structured interviews revealed insights into how emotion mimicry facilitates engaging social interactions in the game.

In summary, this paper contributes by presenting emotion mimicry as a valuable design resource in bodily play. We present EmoEcho, a game that exemplifies this concept, along with a pilot study to examine the user experiences. Our findings highlight the potential of emotion mimicry to enhance the gaming experience, particularly in terms of social engagement. Ultimately, with this work, we hope to inspire further exploration into the integration of emotional elements in digital play, paving the way for more emotionally resonant and socially connected gaming experiences.

2 RELATED WORK

In this section, we will introduce three research areas that influence our game design: mimicry for social engagement, emotion input games and emotional communication for social engagement.

2.1 Mimicry for Social Engagement

Mimicry has been recognized as a powerful tool to enhance social interactions in physical play [27]. Prior research highlights that the ability to sync nonverbally forms a foundation for social connection [32]. Movement synchrony, manifesting in diverse social activities like dancing, playing, and conversing, plays a central role in our everyday social exchanges [12, 31]. In therapeutic settings, movement synchrony between a patient and therapist has been identified as a potential indicator of therapy success [1].

In digital gaming, the social dance game “Yamove!” represents a key example of how synchronizing physical movements can enhance social engagement [18]. This game exemplifies the potential of mimicry in gaming, yet its focus is primarily on movement, leaving emotional aspects relatively unexplored. Robinson et. presented “In the Same Boat”, a game employs synchronized emotional responses as mechanic [33]. In this game, players simultaneously express emotions in response to in-game cues, showing how shared emotional experiences can strengthen social connections in digital play. However, this game only explores synchronization dictated by the game system. Our research diverges from this approach by exploring a different dynamic: one where a player spontaneously expresses an emotion, and the other player then mimics this expression. This shift from system-led to player-driven emotional expression, where one player initiates an emotion and the other follows, offers a new perspective on social interaction in gaming.

2.2 Emotion Input Games

In recent years, there has been an increasing number of games incorporating players’ emotions as inputs. These games are designed to measure and leverage the player’s emotional state, directly influencing various facets of gameplay and, subsequently, shaping the overall gaming experience [8, 10, 21]. Prior works have explored diverse methods for measuring players’ emotions in these games, employing technologies such as Electroencephalography (EEG) [20, 23], facial expression recognition [11, 37], and heart rate sensors [2, 30]. In our research, we opted for facial expression recognition as the chosen method. The rationale behind this choice is twofold. Firstly, EEG-based approaches require specialized

equipment, making them less accessible and practical for players without a technological background, especially in field settings. Secondly, while heart rate sensors are a common choice, establishing an accurate mapping between heart rate variations and specific emotional states presents considerable challenges. Therefore, we adopted facial expression recognition as a starting point for our investigation, aiming to harness its potential in enhancing the emotional dimensions of gameplay.

In early works around emotion input games, researchers explored how players’ affective states can be utilized to adapt the game experience. For instance, Negini et al. [28] introduced an affective game that leveraged players’ affective states to dynamically adjust game characters and the in-game environment, thereby enhancing player performance and overall experience. Similarly, Chanel et al. [4] proposed a strategy to maintain player engagement by tailoring game difficulty based on the player’s emotions. There are also emotion input games designed for emotion regulation. For instance, Bernhaupt et al. [3] developed the game “Sunflower”, which allowed players to control the growth of a virtual flower through their facial expressions. The game encouraged players to express joy, aiming to promote emotion regulation. Another example is Lobel et al.’s “Nevermind”, a horror game that challenged players to regulate their negative emotional states in the face of stressful situations, ultimately enhancing their emotional regulation skills [24]. Similarly, in this research, we investigate using emotion as an input, but with a focus on the social perspective of emotion mimicry in digital play.

2.3 Emotional Communication for Social Engagement

In HCI, prior research has extensively explored the design of emotional communication to enhance social engagement. One approach is the use of public displays of emotion. For example, Howell et al. [16] introduced a thermochromic t-shirt that changes color in response to variations in the wearer’s skin conductance, allowing individuals to express their emotions and engage audiences in shared emotional experiences. Another area of exploration is the development of ambient devices for emotion sharing. The Lumitouch system, for instance, comprises interactive picture frames that use lighting changes to convey emotional content, subtly aiding in maintaining emotional connections over distances [5]. Similarly, the BioCrystal display uses biofeedback to evaluate and communicate the user’s affective states in real-time through an ambient interface [34]. Another example, Breeze, is a wearable pendant that tracks breathing patterns and provides biofeedback, creating intimate channels for emotional exchange [9].

Technological and paradigm advancements have also led to the creation of systems that augment emotional communication. For instance, Neo-Noumena utilizes EEG technology to recognize affective states and project these onto mixed reality platforms, enriching interpersonal communication with additional emotional dimensions [35]. Janssen et al. [19] explored how sharing heart-beat sounds can enhance intimacy and closeness. Liu et al. [22] developed a mobile app for real-time heart rate sharing via text messaging. Hassib et al.’s EmotionActuator system [15] transmits

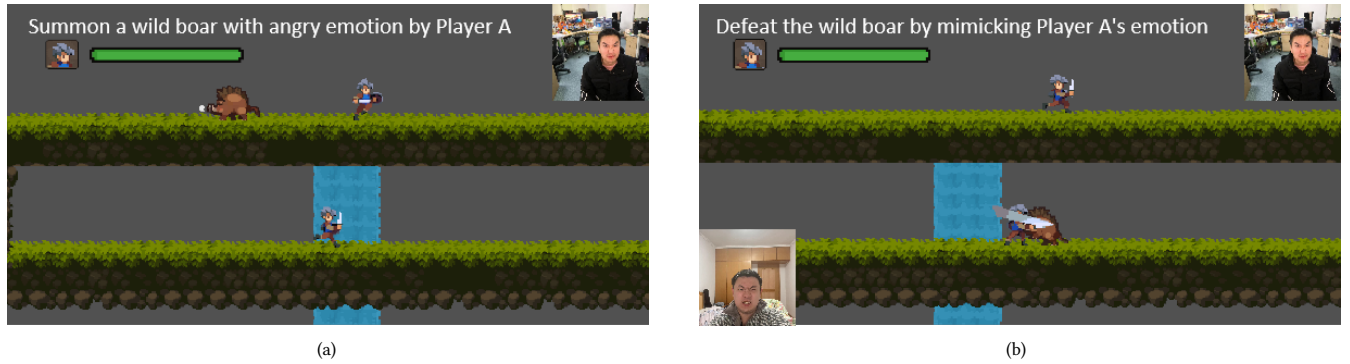


Figure 1: Playing EmoEcho with emotion mimicry. (a) shows Player A summoned a wild boar with angry emotion. (b) shows Player B defeated the wild boar by mimicking A's emotion.

emotional states through electrical muscle stimulation, enabling empathetic understanding of the sender's emotions.

These works demonstrate the potential of designing emotional communication to enhance social engagement. Building on these foundations, our work explores the concept of emotion mimicry as a means to foster social engagement, opening new avenues for empathetic and connected experiences in digital spaces.

3 EMOECHO

EmoEcho is a two-player 2D side-scrolling game that innovatively integrates emotion mimicry into its core gameplay mechanics (Figure 1). This game enables players to dynamically influence the game environment using their facial expressions, creating a unique interactive experience.

3.1 Gameplay

EmoEcho uses facial expressions as the primary input for gameplay. During the game, players are greeted with a horizontally-oriented game environment, initially devoid of interactive content. This sets the stage for a player-generated interactive canvas. The game screen is split horizontally, with the upper half displaying Player A's game area and the lower half showing Player B's. Both areas feature the game map and each player's character. In the top left corner of each player's area, the health status of the character is displayed.

When game starts, both player-controlled characters automatically progress along the track. Player A is given the role of dynamically creating interactive elements within the game—ranging from adversaries to power-ups—by expressing one of the seven identifiable emotions: happiness, fear, anger, disgust, and surprise. Each emotion is linked to different game interactions and elements. Player B, on the other hand, must accurately mimic these facial expressions to interact with the elements—be it utilizing a beneficial item or overcoming an obstacle. This creates a compelling dynamic of emotion mimicry between the players. The detailed design of these interactions, including the specific emotions and their corresponding creations and interactions by Player A and B, are outlined in an accompanying table 1. To facilitate this unique interaction, Player A initiates the gameplay. Player B follows five

seconds later, encountering and responding to the elements generated by Player A. To support the emotion mimicry aspect and to reduce the learning curve [14] for Player B—who might initially struggle to remember the correct facial expression for each interactive element—we have implemented a feature where Player A's expressions are captured in screenshots. These images are then displayed alongside the generated interactive elements. This approach allows Player B to easily see and mimic Player A's expressions, thereby facilitating more seamless and intuitive interactions within the game.

In sum, EmoEcho leverages emotion mimicry as its core gameplay mechanic to facilitate an engaging and socially interactive experience. It expands the boundaries of traditional games that use emotion as input [4, 28], inviting players to delve deep into emotional expressiveness. Additionally, EmoEcho explores new territories beyond traditional physical mimicry in gaming [27], by focusing on emotional mimicry. This approach presents a flexible gaming framework that responds to the players' emotional inputs, allowing for either a competitive race or a cooperative challenge to complete the track together.

3.2 Implementation

EmoEcho was developed using the Unity game development platform. The game runs at a resolution of 1920 x 1080. The track spans a fixed length, designed for a 3-minute gameplay experience. Meanwhile, EmoEcho supports displaying the game on two monitors for gameplay.

EmoEcho uses emotion as game input. We chose to use an external camera, mounted on the monitor, to capture the players' facial expressions continuously in real-time. These expressions are then analyzed by a Convolutional Neural Network (CNN)-based deep learning model, specifically tailored for emotion recognition¹. This model has been pre-trained with extensive facial expression datasets CK+, fine-tuning its ability to discern human emotions accurately. The accuracy rate reaches 94.64% on the CK+ test set.

To maintain fluidity during gameplay, the system is configured to perform facial expression recognition at one-second intervals

¹<https://github.com/WuJie1010/Facial-Expression-Recognition.Pytorch>

Emotion	Game Reaction to Player A's Emotion	Game Reaction to Player B's Mimicry
Happy	Create a health potion	Consume the health potion
Angry	Summon a wild boar enemy	Defeat the wild boar
Fear	Trigger an earthquake	Navigate the quake safely
Disgust	Summon a sticky snail with deceleration debuff	Quickly overcome the snail obstacle
Surprise	Initiate teleportation to a new aesthetic environment	Execute teleportation

Table 1: In-Game Effects Triggered by Emotional Expressions of Players A and B.

aiming to mitigate instances of players inadvertently triggering actions due to insufficient time to switch emoticons. It captures snapshots of the player's expressions, downsizes them to 48x48 pixel grayscale images, and inputs them into the emotion recognition model. The neural network processes these inputs and outputs the recognized emotion, which in turn influences the game. For the integration of neural network inference within the Unity environment, we employed the Barracuda library to ensure minimal latency.

4 PILOT DEPLOYMENT

To evaluate the user experience of EmoEcho, we conducted a comparative study. We developed an alternate version of the game, EmoEcho-keyboard, where game mechanics remained consistent with the original, but the input method differed: players used keyboards to play. We mapped each emotion to a corresponding key on the keyboard, allowing players to execute specific game actions by pressing the corresponding key associated with the desired expression. We recruited two pairs of participants for this study (all males as self-reported, average age $M = 24.5$, $SD = 0.58$). Each pair consisted of individuals who were already friends prior to the study.

The experimental setup included two separate rooms, each equipped with a computer configured to run EmoEcho. Cameras were installed above the monitors to capture the participants' facial expressions. Upon arrival, participants received a brief on the experimental procedure and an explanation of EmoEcho's gameplay mechanics. Following this, we asked participants to practice the range of facial expressions used in the game. Participants then engaged in a trial session, acquainting themselves with both the emotion-based and keyboard-based versions of the game to facilitate ease of operation during the actual study.

Following the trial, we randomly determined the sequence of gameplay conditions for each pair—whether they would engage with the keyboard input or emotion input version first. In this study, the first group started with the emotion input version, while the second group of participants, conversely, started with the keyboard input version. Participants were then directed to the respective rooms for the study. Completion of each condition was followed by participants completing the Core Module and Social Presence Module of the Game Experience Questionnaire (GEQ), which employed a 5-point Likert scale ranging from 'Not at all' to 'Extremely' to gauge aspects of the gaming experience and social presence [17]. Upon conclusion of both gameplay conditions, we conducted semi-structured interviews with the pairs. The interviews centered on their game experiences, e.g., their perceptions of using facial expressions as a control input, and their experiences in mimicking one another's expressions during gameplay.

5 PRELIMINARY USER EXPERIENCE

We present the results from the Game Experience Questionnaire in an accompanying figure (Figure 2) along with our analysis. Participants tended to prefer using facial expressions as game input, as indicated by a higher average score of positive affect in the emotion condition ($M = 3.65$, $SD = 0.41$) compared to the keyboard condition ($M = 1.45$, $SD = 0.93$) and a lower average score of negative affect in the emotion condition ($M = 0.5$, $SD = 0.91$) compared to the keyboard condition ($M = 1.63$, $SD = 0.78$). This preference is also reflected in the lower average score in tension/annoyance using emotion ($M = 0.33$, $SD = 0.27$) than using keyboard ($M = 0.59$, $SD = 0.42$). Regarding the challenges presented by the two input modes, participants found that using a keyboard ($M = 0.19$, $SD = 0.13$) is perceived as easier than using emotion ($M = 1.75$, $SD = 0.74$). Participants reported a distinct sense of flow ($M = 2.45$, $SD = 0.44$), sensory and imaginative immersion ($M = 2.88$, $SD = 0.82$) when using emotion as input. However, when utilizing a keyboard, there is a comparatively diminished perception of flow ($M = 0.7$, $SD = 0.26$), as well as reduced sensory and imaginative immersion ($M = 1$, $SD = 0.65$). Participants generally felt a sense of competence in both conditions of using emotion ($M = 3.05$, $SD = 0.66$) and using a keyboard ($M = 2.6$, $SD = 0.23$). The higher average score in behaviour involvement ($M = 2.88$, $SD = 0.72$) and empathy ($M = 3.21$, $SD = 0.8$) while using emotion and the lower average score in behaviour involvement ($M = 0.75$, $SD = 0.44$) and empathy ($M = 0.79$, $SD = 0.46$) while using a keyboard indicates the participants engaged socially and had emotional connections with each other when one's facial expression is imitated by the other one. The participants felt almost no negative feelings in both conditions: emotion ($M = 0.7$, $SD = 0.2$) and keyboard ($M = 0.7$, $SD = 0.70$).

During our interviews, all the participants expressed appreciation for using facial expressions as game input. This unique approach provided them with an increased sense of immersion, a feeling they reported as unattainable with traditional keyboard-based games. One participant shared: *"I tried the emotion input version first and it was incredibly novel and fun. I made various expressions to create obstacles for my co-player. It felt like I was becoming the master of the game world."*

Regarding emotion mimicry, all participants conveyed positive feedback. They believed that mimicking emotions facilitated the rapid establishment of a close connection with their game partner. Participants noted that while mimicking facial expressions, they could better understand and feel the other person's emotions. For instance, a participant said: *"I think the emotion mimicry is pretty cool. While trying to copy my co-player's expressions, I experienced a moment where my actual feelings were influenced. Like when I imitated an angry face, I briefly felt a bit mad, but then quickly*

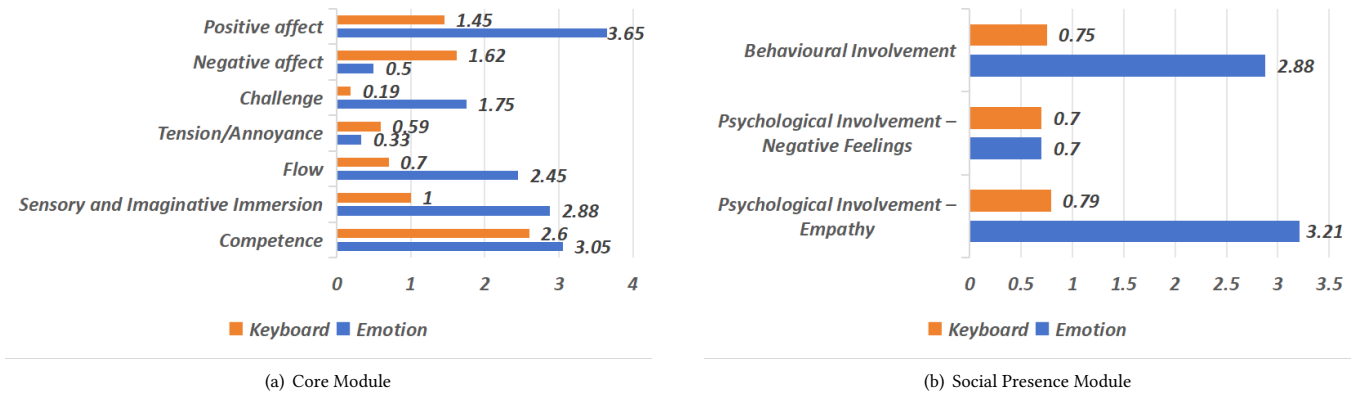


Figure 2: Average component scores from Game Experience Questionnaire

returned to feeling happy. I also felt more attuned to my co-player's mood changes during the game, almost like I was empathizing with him."

Participants' feedback indicated that they have the freedom to choose between cooperating to complete the game together or making it difficult for the other during the gaming process, and this choice fluctuated as the game progresses. One participant stated: "I would initially set up some obstacles for him, and when he is close to failing, I would summon some blood potions to help him pass the level. I hope to increase the difficulty of his gaming experience while ensuring that he can still pass the level."

Overall, our initial user study has given us valuable insights into the effects of emotion mimicry on game experience and social presence. The results above suggest that our designed game, EmoEcho, not only immerses players in using facial expressions for gameplay but also improves the overall gaming experience and fosters interpersonal relationships through the emotion mimicry mechanism.

6 FUTURE WORK

For the next step, we plan to iterate the game and continue with the user study. Our preliminary study has shown positive player experience. We will involve more game designers specializing in emotional design to evaluate whether the game design optimally utilizes emotional mimicry. Iterations will be made based on their feedback. Subsequently, we aim to invite a total of 32 participants, comprising 16 pairs, to interact with our game EmoEcho. We will conduct statistical analysis to examine the quantitative data, coupled with thematic analysis to extract meaningful insights from the interview data. By deeply reflecting on both the design process and the study outcomes, we will present a set of design guidelines. These guidelines will focus on designing emotion mimicry for facilitating engaging social play experiences.

Over a longer term, we hope to address the following issues. Firstly, our current questionnaire can only reflect general information, but for pinpointing which specific gaming elements or mechanisms contribute to a better gaming experience and enhanced social

presence, we need more cases about emotion mimicry to study. Secondly, the extended engagement with emotion mimicry as a game mechanism may cause facial muscle fatigue. While this concern was not raised by any participants in our pilot study, this might be because the duration of the study was limited. If the game were to be deployed in the field, the uncontrollable game duration could potentially impose excessive strain on the players' facial muscles. We will continue researching the enduring playability of the emotion mimicry, seeking to address potential problems related to facial muscle fatigue in prolonged gameplay. Thirdly, requiring players in a specific emotional state to make facial expressions inconsistent with their current mood can lead to negative feelings. Our work will explore design strategies to synchronize players' facial expressions with both the game content and their current emotional states in the future.

7 CONCLUSION

In conclusion, our research delves into the use of emotion mimicry as a key mechanic in digital social games. This innovative approach is rooted in the well-established field of bodily play, highlighting how movement mimicry can facilitate social engagement. To illustrate this concept, we developed "EmoEcho", a two-player game that requires players to mimic the other's facial expressions as a means of interaction. A pilot study was conducted to evaluate the game's impact. The initial results indicated a positive user experience and emphasized the role of emotion mimicry in fostering a sense of togetherness among participants. Looking ahead, this study aims to inspire designers to integrate emotion mimicry into their future interactive designs, recognizing its potential to enhance both the emotional depth and social connectedness of digital experiences.

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